LINEAR ANTI-RECURRENCE SEQUENCES ARE LINEAR PLUS AUTOMATIC

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ABSTRACT. Let $\mathbf{a} = (a_1, \dots, a_k)$ be a vector of positive integers, $k \geq 2$, with trace $\tau = \sum a_j$. Its anti-recurrence sequence is the unique increasing sequence (A_n) with $A_n = \sum_j a_j B_{k(n-1)+j}$, where (B_m) enumerates the complement of $\{A_n\}$ in $\mathbb{N}$; for $\mathbf{a} = (1,1)$ it is the anti-Fibonacci sequence (OEIS A075326). Fokkink and Joshi conjectured that $A_n - \kappa n$ is τ-automatic, where $\kappa = k\tau + 1$, and proved this under a restriction on $\mathbf{a}$ that fails, for example, for the "rusty numbers" $\mathbf{a} = (1, d)$, $d \geq 3$. We prove the conjecture for every $\mathbf{a}$ in two steps. First we prove its weak form, $-k\tau \leq A_n - \kappa n \leq \tau - k$. Then we read off from this proof a two-term recursion $d(\tau q + r + 1) = G_r(d(q), d(q+1))$ for $d(n) = A_n - \kappa n$, which gives an explicit τ-automaton with at most $1 + (k\tau + \tau - k + 1)^3$ states. For $\mathbf{a} = (1,4)$ we give the minimal automaton, an independent Walnut proof, and an explanation of why the progression $A_{5q+1} = 9 + 55q$ observed by Fokkink and Joshi first fails at A_{1741}. We conjecture that the box criterion of the first step always gives the exact range of $A_n - \kappa n$, and support this with exact computations on 746 forms.

1. INTRODUCTION

1.1. **The result in brief.** Fix positive integers $a_1, \dots, a_k$ with $k \geq 2$. Repeat the following step: take the k smallest positive integers $b_1 < \cdots < b_k$ that have not been used yet, and mark them as used; then put $A = a_1 b_1 + \cdots + a_k b_k$ into the sequence and mark A as used too. The numbers A produced in this way, $A_1 < A_2 < \cdots$, form the *anti-recurrence sequence* of $\mathbf{a} = (a_1, \dots, a_k)$ [5, 6]. For $\mathbf{a} = (1,1)$ the rule gives the *anti-Fibonacci numbers* $3, 9, 13, 18, 23, 29, \dots$ (OEIS A075326 [16], which starts with an extra 0). Fokkink and Joshi conjectured that these sequences are always linear plus automatic. We prove this.

Main Theorem. *Let $\tau = a_1 + \cdots + a_k$ and $\kappa = k\tau + 1$. Then $A_n = \kappa n + d(n)$, where d is a τ-automatic sequence and $-k\tau \leq d(n) \leq \tau - k$ for all $n \geq 1$. In particular, the conjecture of Fokkink and Joshi (Conjecture 1.1) holds for every positive integral linear form.*

It combines Theorems A and C below; Theorem B gives an explicit recursion for d.

1.2. **Background.** Formally, the anti-recurrence sequence of $\mathbf{a}$ is the pair of complementary, strictly increasing sequences $(A_n)_{n \geq 1}$, $(B_m)_{m \geq 1}$ in $\mathbb{N} = \{1, 2, \dots\}$ such that

$$A_n = \sum_{j=1}^{k} a_j \, B_{k(n-1)+j} \qquad (n \geq 1). \tag{1}$$

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Such a pair exists and is unique [5, Lemma 3], [6, Lemma 2], and B is generated greedily by the mex rule, as described above. Fokkink and Joshi call $\mathbf{a}$ a positive integral linear form of dimension k, and τ its *trace*.

The anti-Fibonacci sequence was entered in the OEIS by A. Murthy in 2002; the mex definition is due to P. Lallouet (2008) and the name to D. R. Hofstadter (2014) [16, 9]. Zaslavsky [23, Thm. 1] gave a closed formula in terms of "utterly odd" numbers.[1] Equivalently, $A_n = 5n - 2 + \mathrm{PD}_{n-1}$, where $\mathrm{PD} = 0, 1, 0, 0, 0, 1, 0, 1, \ldots$ is the period-doubling sequence (OEIS A096268, indexed from 0) [5, eq. (2)], [6, §1]; Bosma et al. [3, §3] verified this with Walnut. Kimberling and Moses [12] placed the example in a theory of systems of complementary equations and conjectured substitutions for the anti-m-nacci systems; Bosma et al. [3] proved the anti-tribonacci and anti-tetranacci cases with the theorem prover Walnut [15] and formulated the *Clergyman's Conjecture* for anti-k-naccis. Fokkink and Joshi [5, 6] then introduced general linear forms and conjectured:

Conjecture 1.1 (Fokkink–Joshi [5, Conj. 4], [6, Conj. 2])**.** *For every positive integral linear form* $\mathbf{a}$ *of dimension* $k > 1$*, the sequence* $A_n - \kappa n$ *is* τ*-automatic.*

(It is Conjecture 4 in arXiv v1 and Conjecture 2 in the published version.) Both versions call the statement "$A_n - \kappa n$ is bounded" the *weak form* of the conjecture, and note that Kimberling and Moses phrase their conjectures in this form. Fokkink and Joshi proved Conjecture 1.1 for the anti-k-bonaccis $\mathbf{a} = (1, \ldots, 1)$ [5, Thm. 8], [6, Thm. 3]; for A_1*-bounded* forms, $A_1 \le (k-1)\tau + 2$ [5, Thm. 12], [6, Thm. 4]; and, with Walnut, for the anti-Pell form $(1, 2)$ and the anti-Jacobsthal form $(2, 1)$. For $k = 2$, A_1-boundedness means $a_2 \le 2$.

For $\mathbf{a} = (1, d)$ they call the A_n the *d-rusty numbers*. For $d = 3$, which is not A_1-bounded, they display a guessed base-4 automaton with 6 states [6, Fig. 7]. For $d = 4$ the sequence begins $9, 19, 29, 39, 54, 64, 74, 84, 98, 109, \ldots$, and they observe that $A_{5n+1} = 9 + 55n$ holds "up until $n = 348$, when $A_{1741} \neq 9 + 55 \cdot 348$"; "proving or disproving the conjecture for the rusty numbers remains a challenge" [6, §5]. Their final section asks: "Does the conjecture hold without the restriction of A_1-boundedness? Are the rusty numbers sums of linear sequences and automatic sequences?" [6, §6]. The Main Theorem answers both questions affirmatively.

1.3. **Strategy: first the weak conjecture, then an automaton.** The proof has two steps. Throughout, $d(n) := A_n - \kappa n$ for $n \ge 1$ and $d(0) := 0$ (the convention $A_0 = 0$); further notation is collected in Table 1.

Step 1: the weak conjecture (Section 3). A counting identity (Lemma 3.1) turns (1) into a self-referential formula that expresses $d(n)$ through the number of earlier values $d(i)$ below certain thresholds. An induction then shows that every integer interval $[L, U]$ (a *box*) satisfying four explicit inequalities contains all values of d (Proposition 3.2), and a short estimate shows that $[-k\tau, \tau - k]$ is such a box (Lemma 3.4).

Theorem A (Boundedness)**.** *For every positive integral linear form of dimension* $k \ge 2$ *and every* $n \ge 1$*,*

$$-k\tau \;\le\; A_n - \kappa n \;\le\; \tau - k.$$

More generally, every box satisfying conditions (i)–(iv) of Proposition 3.2 contains all $d(n)$*.*

Step 2: an automaton from the proof (Section 4). Once d is known to lie in $[-k\tau, \tau - k]$, all but two terms of each count in the formula of Step 1 are determined. What remains is a recursion in which $d(\tau q + r + 1)$ depends only on $d(q)$ and $d(q + 1)$:

[1] Read literally with the definition of "utterly odd" in [23], the two odd cases of [23, Thm. 1] are interchanged (in his indexing $\bar{F}_0 = 3$, and $\bar{F}_1 = 9 = 5 \cdot 1 + 4$ although 0 is not utterly odd). The corrected reading agrees with the terms listed in [23]; we checked it against the first 3000 terms.

Theorem B (Two-term recursion)**.** *For $0 \le r < \tau$ and $y_0, y_1 \in \mathbb{Z}$ put*

$$G_r(y_0, y_1) = A_1 - K - 1 - r - \tau + \sum_{j=1}^{k} a_j \Big([\, y_0 \le kr + j - 1 \,] + [\, y_1 \le kr + j - 1 - K \,] \Big),$$

where $K = k\tau$, $A_1 = \sum_j j a_j$, and $[\,P\,] \in \{0,1\}$ is the Iverson bracket. Then $d(\tau q + r + 1) = G_r\big(d(q), d(q+1)\big)$ for all $q \ge 0$ and $0 \le r < \tau$.

Triples of consecutive values of d then serve as the states of an automaton:

Theorem C (Automaticity)**.** *$(d(n))_{n \ge 0}$ is τ-automatic. It is generated by an explicit msd-first τ-DFAO with at most $1 + (k\tau + \tau - k + 1)^3$ states. In particular Conjecture 1.1, and hence its weak form, holds for all positive integral linear forms.*

Section 3.3 states the sharp-box conjecture, Section 5 treats $\mathbf{a} = (1, 4)$, and Section 6 discusses related work and further directions. The appendices contain the computational checks, further Walnut certificates and a variant of Step 2 that works with any box.

1.4. **A worked example: the anti-Fibonacci numbers.** For $\mathbf{a} = (1, 1)$ we have $k = \tau = 2$, $K = 4$, $\kappa = 5$ and $A_1 = 3$. The greedy rule gives $B = 1, 2, 4, 5, 6, 7, 8, 10, 11, 12, \ldots$ and

n	1	2	3	4	5	6	7	8
A_n	3	9	13	18	23	29	33	39
$d(n) = A_n - 5n$	-2	-1	-2	-2	-2	-1	-2	-1

Step 1. Theorem A gives $-4 \le d(n) \le 0$. The narrower box $[L, U] = [-2, -1]$ also satisfies the criterion of Proposition 3.2: the function Φ defined there takes the values $\Phi(0,-2) = -2$, $\Phi(1,-2) = -1$ and $\Phi(0,-1) = \Phi(1,-1) = -2$, so (i) and (ii) hold, and (iii) $U \ge -k(\tau - 1) = -2$ and (iv) $L \le 0$ are clear. Hence $d(n) \in \{-2, -1\}$ for all $n \ge 1$.

Step 2. Inserting $d \in \{-2, -1\}$ (and $d(0) = 0$) into Theorem B, all Iverson brackets but one are constant, and the recursion becomes

$$d(2m + 1) = -2 \quad (m \ge 0), \qquad d(2m) = -3 - d(m) \quad (m \ge 1).$$

Reading the binary digits of n from the most significant end, this is a 3-state automaton: a start state s with output $0 = d(0)$, a state α with output -2 and a state β with output -1, with transitions $s \xrightarrow{0} s$, $s \xrightarrow{1} \alpha$, $\alpha \xrightarrow{0} \beta$, $\alpha \xrightarrow{1} \alpha$ and $\beta \xrightarrow{0,1} \alpha$. Since $\mathrm{PD}_{2m} = 0$ and $\mathrm{PD}_{2m+1} = 1 - \mathrm{PD}_m$, the sequence $-2 + \mathrm{PD}_{n-1}$ satisfies the same recursion, which recovers the formula $A_n = 5n - 2 + \mathrm{PD}_{n-1}$ recalled above. For general $\mathbf{a}$ the same two steps apply verbatim; only the box and the automaton become larger.

1.5. **The 4-rusty numbers.** For $\mathbf{a} = (1, 4)$ ($\tau = 5$, $\kappa = 11$), d takes exactly the values $-6, \ldots, 1$. From Theorem B we deduce that $A_{5q+1} = 9 + 55q$ if and only if $d(q) \ne 1$, and that the first such q is 348, reached by the chain $3 \to 14 \to 70 \to 348$ (Corollary 5.1). The minimal automaton has 17 states in our convention and 9 in that of [6], and Walnut certifies directly, without using Theorems A–C, that it generates the 4-rusty numbers (Theorem 5.2).

2. Notation and preliminaries

A *box* is a pair of integers $L \le U$ satisfying (i)–(iv) of Proposition 3.2. The first block is $B_j = j$ for $j \le k$, so $A_1 = \sum_j j a_j$ and $d(1) = A_1 - K - 1$. Moreover

$$\tau \;\le\; A_1 \;\le\; K - (k - 1), \tag{2}$$

since $K - A_1 = \sum_j (k - j) a_j \ge \sum_{j<k} (k - j) \ge k - 1$. We use the gap bound $A_{n+1} - A_n \ge K$ [5, Lemma 2], [6, Lemma 1]; in fact only $A_{n+1} - A_n \ge 2$ is needed.

TABLE 1. Notation.

$\mathbf{a} = (a_1, \dots, a_k)$	the linear form; $k \geq 2$, all $a_j \geq 1$
$\tau = \sum_j a_j$	trace; the base of the automata
$K = k\tau, \quad \kappa = K + 1$	κ is the slope of A_n
$A_1 = \sum_j j a_j$	first term of A
$d(n) = A_n - \kappa n$	discrepancy; $d(0) = 0$
$c(m) = \#\{i : A_i < B_m\}$	counting function (Lemma 3.1)
$n - 1 = \tau q + r, \quad u_j = kr + j - 1$	index decomposition, $0 \leq r < \tau$, $1 \leq j \leq k$
$\Phi(r, x)$	box function (Section 3.2)
$G_r(y_0, y_1)$	two-term recursion (Theorem B)
$V(m) = (d(m-1), d(m), d(m+1))$	automaton state (Section 4)
$[\,P\,]$	Iverson bracket: 1 if P holds, else 0

A sequence $(s(n))_{n\geq 0}$ is *τ-automatic* if there is a deterministic finite automaton with output (DFAO) that, reading the base-τ representation of n, ends in a state whose output is $s(n)$. Our automata read the most significant digit first and allow leading zeros, as Walnut does; they handle leading zeros explicitly (Theorem C). The reading direction does not affect the class of automatic sequences, and s is τ-automatic if and only if its *τ-kernel* $\{(s(\tau^e n + b))_{n\geq 0} : e \geq 0,\ 0 \leq b < \tau^e\}$ is finite; see [1, Chs. 5–6] and [4]. We use the kernel characterisation only in Appendix C.

3. Step 1: boundedness (the weak conjecture)

3.1. **A counting identity.** For $m \geq 1$ let $c(m) := \#\{i \geq 1 : A_i < B_m\}$.

Lemma 3.1.

(a) *$B_m = m + c(m)$. Moreover $A_i < B_m$ if and only if $Ki + d(i) \leq m - 1$.*
(b) *Let $n \geq 1$ and $m_j := k(n-1)+j$ for $1 \leq j \leq k$. Then no index $i \geq n$ satisfies $A_i < B_{m_j}$.*
(c) *Write $n - 1 = \tau q + r$ with $0 \leq r < \tau$, and put $u_j := kr + j - 1 \in [0, K-1]$. Then $m_j - 1 = Kq + u_j$, and*

$$d(n) = A_1 - K - n + \sum_{j=1}^{k} a_j\, c(m_j), \qquad c(m_j) = \#\{i \geq 1 :\ d(i) \leq K(q-i) + u_j\}. \tag{3}$$

Proof. (a) The integers in $[1, B_m)$ are the $m-1$ numbers $B_{m'}$ with $m' < m$, together with the $c(m)$ numbers $A_i < B_m$. Hence $B_m - 1 = (m-1) + c(m)$. Since A is strictly increasing, exactly $A_i - i$ of the B's lie below A_i. So $A_i < B_m$ if and only if $A_i - i \leq m - 1$, and $A_i - i = Ki + d(i)$.

(b) Because $k \geq 2$ and every $a_j \geq 1$, we have $A_n \geq B_{kn} + B_{kn-1} > B_{kn}$. So at least kn of the B's lie below A_n, and $A_n - n \geq kn \geq m_j > m_j - 1$. Since $A_{i+1} - A_i \geq 2$, the quantity $A_i - i$ is strictly increasing in i. So $A_i - i > m_j - 1$ for every $i \geq n$, and (a) applies.

(c) By (1) and (a), $A_n = \sum_j a_j(m_j + c(m_j))$, and $\sum_j a_j m_j = K(n-1) + A_1$. Subtracting $\kappa n = (K+1)n$ gives the first formula. The second is (a) with $m_j - 1 = Kq + u_j$. As (r, j) ranges over $[0, \tau - 1] \times [1, k]$, the number u_j takes each value in $[0, K-1]$ exactly once. □

3.2. **Invariant boxes.** For $0 \leq r < \tau$ and $x \in \mathbb{Z}$ define

$$\Phi(r, x) := A_1 - K - 1 - r + \sum_{j=1}^{k} a_j \Big\lfloor \frac{kr + j - 1 - x}{K} \Big\rfloor.$$

Proposition 3.2 (Box criterion)**.** *Let $L \leq U$ be integers with*

(i) $\Phi(r, L) \le U$ *for all* r*;*
(ii) $\Phi(r, U) \ge L$ *for all* r*;*
(iii) $U \ge -k(\tau - 1)$*;*
(iv) $L \le 0$.

Then $L \le d(n) \le U$ *for all* $n \ge 1$.

Proof. We use strong induction on n, including the base case $n = 1$ (where the hypothesis is empty). Let $n = \tau q + r + 1$. By Lemma 3.1(b), $c(m_j) = \#\{1 \le i < n : Ki \le Kq + u_j - d(i)\}$.

Upper bound. By induction $d(i) \ge L$ for $i < n$, so $c(m_j) \le \#\{i \ge 1 : Ki \le Kq + u_j - L\}$. By (iv), $Kq + u_j - L \ge 0$. Hence the count is $\lfloor (Kq + u_j - L)/K \rfloor = q + \lfloor (u_j - L)/K \rfloor$.

Lower bound. By induction $d(i) \le U$ for $i < n$. So every $i \ge 1$ with $i < n$ and $Ki \le Kq + u_j - U$ is counted, i.e. every $i < n$ with $1 \le i \le q + \lfloor (u_j - U)/K \rfloor$. We claim that all such i satisfy $i < n$.

- By (iii), $u_j - U \le (K - 1) + (K - k) < 2K$, so $i \le q + 1$.
- If $q \ge 1$, then $q + 1 \le \tau q < n$.
- If $q = 0$ and $r \ge 1$, then $i \le 1 < n$.
- If $q = r = 0$, then $u_j \le k - 1$, and (iii) gives $u_j - U \le K - 1$, so no $i \ge 1$ qualifies.

Hence $c(m_j) \ge q + \lfloor (u_j - U)/K \rfloor$.

Inserting both bounds into (3), with $n = \tau q + r + 1$ and $\sum_j a_j q = \tau q$, gives $\Phi(r, U) \le d(n) \le \Phi(r, L)$. Then (i) and (ii) close the induction. □

Put $c_0 := A_1 - K - 1 + (A_1 - \tau)/K$. We have $\sum_j a_j \, kr/K = r$ and $\sum_j a_j (j - 1) = A_1 - \tau$. Summing $y - 1 < \lfloor y \rfloor \le y$ with weights a_j therefore gives

$$c_0 - \frac{x}{k} - \tau \;<\; \Phi(r, x) \;\le\; c_0 - \frac{x}{k} \qquad (0 \le r < \tau,\ x \in \mathbb{Z}). \tag{4}$$

Since $k \ge 2$, the map $x \mapsto c_0 - x/k$ is a contraction, so boxes exist. Explicitly:

Proposition 3.3. *Let*

$$L^* = \min\left\{0,\ \left\lfloor \tfrac{c_0(1-1/k)-\tau+1}{1-1/k^2} \right\rfloor,\ \left\lfloor k(c_0 - 1) + k^2(\tau - 1) \right\rfloor,\ \left\lfloor \tfrac{k(c_0-1)}{k+1} \right\rfloor \right\}$$

and $U^* = \lfloor c_0 - L^*/k \rfloor$*. Then* (L^*, U^*) *satisfies (i)–(iv) and* $L^* \le U^*$.

Proof. (i) By (4), $\Phi(r, L^*) \le c_0 - L^*/k$, and Φ is an integer, so $\Phi(r, L^*) \le U^*$.

(ii) By (4), $\Phi(r, U^*) > c_0 - U^*/k - \tau \ge c_0(1 - 1/k) + L^*/k^2 - \tau$. The right-hand side is $\ge L^* - 1$ by the second entry of the minimum. Since Φ is an integer, $\Phi(r, U^*) \ge L^*$.

(iii) We have $U^* > c_0 - L^*/k - 1 \ge -k(\tau - 1) - 1$ by the third entry; U^* is an integer, so $U^* \ge -k(\tau - 1)$.

$L^* \le U^*$: similarly, $U^* > c_0 - L^*/k - 1 \ge L^* - 1$ by the fourth entry.

(iv) holds because 0 is in the minimum. □

This already proves the weak form of Conjecture 1.1. The following uniform box is sharper, and it is what makes Step 2 work.

Lemma 3.4 (Universal box)**.** *Let* $L = -K$ *and* $U = \max\{-k(\tau - 1),\ \max_r \Phi(r, -K)\}$*. Then (i)–(iv) hold,* $L < U$*, and* $U \le \tau - k$*. Consequently* $-K \le d(n) \le \tau - k$ *for all* $n \ge 1$*, and also for* $n = 0$ *since* $d(0) = 0$ *and* $\tau \ge k$.

Proof. Conditions (i), (iii) and (iv) hold by construction. Also $L = -K < -K + k = -k(\tau - 1) \le U$.

$U \le \tau - k$. By (4) and (2), using $(A_1 - \tau)/K < 1$,

$$\Phi(r, -K) \le c_0 + \tau < A_1 - K + \tau \le \tau - k + 1.$$

Since Φ is an integer, $\Phi(r,-K) \le \tau - k$. Also $-k(\tau-1) \le 0 \le \tau - k$.

Condition (ii). By (4), $\Phi(r,U) > c_0 - U/k - \tau$. By (2), $c_0 \ge \tau - K - 1$.

- If $U = \max_r \Phi(r,-K)$, then $U \le c_0 + \tau$, and since $1 - 1/k > 0$,

$$\Phi(r,U) > c_0\Big(1-\frac{1}{k}\Big) - \tau\Big(1+\frac{1}{k}\Big) \ge (\tau - K - 1)\Big(1-\frac{1}{k}\Big) - \tau\Big(1+\frac{1}{k}\Big) = -K - 1 + \tau\Big(1-\frac{2}{k}\Big) + \frac{1}{k}.$$

  The right-hand side exceeds $-K-1$.
- If $U = -k(\tau-1)$, then $\Phi(r,U) > c_0 + (\tau-1) - \tau \ge \tau - K - 2 \ge -K - 1$.

In both cases the integer $\Phi(r,U)$ exceeds $-K-1$, so $\Phi(r,U) \ge -K = L$. □

Proof of Theorem A. Combine Lemma 3.4 with Proposition 3.2. □

3.3. **Sharp boxes: a conjecture.** By Proposition 3.2, every box satisfying (i)–(iv) contains the range of d. For fixed $L \le 0$ the least admissible U is $\max\{-k(\tau-1), \max_r \Phi(r,L)\}$, so a *narrowest* box (one minimising $U - L$) can be found by a one-dimensional search over L. The *exact* range of d can be computed from the automaton of Theorem C, as the set of outputs of its reachable states. Finite prefixes are not reliable for this, since extreme values can first occur very late: for $(200, 1)$ the exact minimum is -400, while the first $3 \cdot 10^6$ terms stay above -370.

Conjecture 3.5 (Sharp boxes)**.** *For every positive integral linear form, the narrowest box satisfying (i)–(iv) of Proposition 3.2 is exactly* $[\min_n d(n), \max_n d(n)]$.

In other words, the criterion of Step 1 would determine the range of $A_n - \kappa n$ exactly, not just bound it. As evidence, we compared the two exactly (without sampling any n) on the 10 named forms $(1,1)$, $(1,2)$, $(2,1)$, $(1,3)$, $(1,4)$, $(1,5)$, $(2,7)$, $(1,1,1)$, $(1,1,1,1)$, $(1,1,5)$ and on 736 further forms: all 400 forms with $k = 2$ and $a_j \le 20$, all 216 forms with $k = 3$ and $a_j \le 6$, and 120 random forms with $4 \le k \le 6$ and $a_j \le 12$. In all 746 cases the narrowest box equals the exact range; for example, both are $[-2,-1]$ for $(1,1)$ (Section 1.4) and $[-6,1]$ for $(1,4)$. An independently written second implementation (Appendix A) found the same on 1682 forms.

Remark 3.6. (a) *The explicit box.* The box (L^*, U^*) of Proposition 3.3 can exceed the narrowest box by up to 2 units of total width. For $(1,2)$ it is $[-4,0]$, while the exact range is $[-3,-1]$. In the 736-form sweep the excess is 0, 1, 2 for 49, 277, 410 forms respectively, and never more than 2.

(b) *Sharpness of Lemma 3.4.* The lower end of Lemma 3.4 is nearly sharp: in every tested form $\min d \ge -K + 2$, and in our 736-form sweep equality holds exactly for the 20 forms $(a,1)$, $1 \le a \le 20$ (for example $(1,1)$ and $(2,1)$). We have not proved $d \ge -K+2$ in general. The upper end $\tau - k$ is far from sharp; the second implementation found $\max d \le 0.66(\tau - k)$ on all its forms.

4. Step 2: from the proof to an automaton

Proof of Theorem B. Let $n = \tau q + r + 1$ and apply (3). Write $i = q - t$.

- Every $i \le q - 1$ ($t \ge 1$) is counted, because by Lemma 3.4, $d(i) \le \tau - k < K \le Kt + u_j$.
- No $i \ge q + 2$ is counted, because $K(q-i) + u_j \le -2K + K - 1 < -K \le d(i)$.

Hence, for $q \ge 1$,

$$c(m_j) = (q-1) + [\, d(q) \le u_j \,] + [\, d(q+1) \le u_j - K \,].$$

For $q = 0$ the count is $c(m_j) = [\, d(1) \le u_j - K \,]$. This agrees with the same formula, because $d(0) = 0 \le u_j$. Substituting into (3), with $\sum_j a_j(q-1) = \tau q - \tau$ and $n = \tau q + r + 1$, gives $d(n) = G_r(d(q), d(q+1))$. □

Proof of Theorem C. For $m \geq 1$ let $V(m) := (d(m-1), d(m), d(m+1)) \in \mathbb{Z}^3$. By Lemma 3.4 (including $d(0) = 0$), $V(m) \in [-K, \tau - k]^3$ for all $m \geq 1$. Fix a digit $0 \leq c < \tau$ and $m \geq 1$. The coordinates of $V(\tau m + c)$ are $d(\tau m + c + e)$ for $e \in \{-1, 0, 1\}$. Write $\tau m + c + e = \tau q' + r' + 1$ with $0 \leq r' < \tau$.

- If $c + e - 1 \geq 0$, then $q' = m$ and $r' = c + e - 1$.
- If $c + e - 1 \in \{-2, -1\}$, then $q' = m - 1$ and $r' = \tau + c + e - 1$, which lies in $[0, \tau - 1]$ because $\tau \geq 2$.

In either case $q' \geq 0$. By Theorem B, each coordinate is $G_{r'}$ applied to two consecutive entries of $V(m)$. So there are maps $H_c : \mathbb{Z}^3 \to \mathbb{Z}^3$, depending only on c and $\mathbf{a}$, with

$$V(\tau m + c) = H_c(V(m)) \qquad (m \geq 1,\ 0 \leq c < \tau). \tag{5}$$

Define a DFAO as follows.

- States: a start state S, plus the vectors $V(m)$ for $m \geq 1$. These lie in $\{S\} \cup [-K, \tau - k]^3$, which has at most $1 + (K + \tau - k + 1)^3$ elements.
- Transitions: $S \xrightarrow{0} S$; $S \xrightarrow{c} V(c)$ for $1 \leq c < \tau$; and $V \xrightarrow{c} H_c(V)$.
- Outputs: S outputs $0 = d(0)$; a vector outputs its middle coordinate.

Feed in the base-τ digits of n, most significant first, with leading zeros allowed. By (5) and induction on the number of digits after the leading zeros, the automaton ends in S if $n = 0$ and in $V(n)$ otherwise. So it outputs $d(n)$. The sequence $(A_n - \kappa n)_{n \geq 1}$ is the restriction of $(d(n))_{n \geq 0}$ to $n \geq 1$, and the same automaton computes it. □

The argument does not depend on the specific box of Lemma 3.4: Appendix C gives a version that works with any box satisfying (i)–(iv), via the τ-kernel.

5. The 4-rusty numbers

Here $\mathbf{a} = (1, 4)$: $k = 2$, $\tau = 5$, $K = 10$, $\kappa = 11$, $A_1 = 9$. With $u_1 = 2r$, $u_2 = 2r + 1$ and $j = r + 1$, Theorem B becomes

$$d(5q+j) = -j - 6 + [\, d(q) \leq 2j-2\,] + 4[\, d(q) \leq 2j-1\,] + [\, d(q+1) \leq 2j-12\,] + 4[\, d(q+1) \leq 2j-11\,] \tag{6}$$

for $q \geq 0$ and $1 \leq j \leq 5$. Conditions (i)–(iv) of Proposition 3.2 hold for $[L, U] = [-6, 1]$. The range of d is exactly $[-6, 1]$ (the output set of the automaton below), so this is the narrowest possible box. Evaluating (6) with $d \in [-6, 1]$ gives, for all $q \geq 0$:

$$\begin{aligned}
d(5q+1) &= -3 + [\, d(q) \leq 0\,], \qquad d(5q+2) = -3,\\
d(5q+3) &= -4 + [\, d(q+1) \leq -6\,] + 4[\, d(q+1) \leq -5\,],\\
d(5q+4) &= -5 + [\, d(q+1) \leq -4\,] + 4[\, d(q+1) \leq -3\,],\\
d(5q+5) &= -6 + [\, d(q+1) \leq -2\,] + 4[\, d(q+1) \leq -1\,].
\end{aligned}$$

In particular $A_{5q+2} = 19 + 55q$ for every $q \geq 0$.

Corollary 5.1. *$A_{5q+1} = 9 + 55q$ if and only if $d(q) \neq 1$. The least q with $d(q) = 1$ is $q = 348$. Hence $A_{5q+1} = 9 + 55q$ for $q < 348$, and $A_{1741} = 19148 \neq 9 + 55 \cdot 348$.*

Proof. $A_{5q+1} = 9 + 55q$ means $d(5q + 1) = -2$, i.e. $d(q) \leq 0$, i.e. $d(q) \neq 1$. From the displayed formulas:

- $d(n) = 1$ if and only if $n = 5q + 3$ with $d(q + 1) = -6$;
- $d(n) = -6$ if and only if $n = 5p$ with $d(p) \geq 0$;
- $d(n) \geq 0$ if and only if either $n = 5q + 4$ with $d(q + 1) \leq -4$, or $n = 5q + 3$ with $d(q + 1) \leq -5$.

Since $d(1), d(2), d(3), d(4) = -2, -3, -4, -5$, the least n with $d(n) \geq 0$ is $5 \cdot 2 + 4 = 14$. Then the least n with $d(n) = -6$ is $5 \cdot 14 = 70$, and the least n with $d(n) = 1$ is $5 \cdot 69 + 3 = 348$. In base 5 the chain is $3 \to 24 \to 240 \to 2343$, and the first failure is at index $5 \cdot 348 + 1 = 1741 = (23431)_5$. The value $A_{1741} = 11 \cdot 1741 - 3 = 19148$ follows from $d(1741) = -3$. □

TABLE 2. Minimal msd-first 5-DFAO for $d(n) = A_n - 11n$, $\mathbf{a} = (1, 4)$, in our convention (input $n \geq 0$, leading zeros allowed, state 0 is initial, $d(0) = 0$). This is the Walnut word automaton `RUS`, whose stored outputs are $d + 6$.

state	out	0	1	2	3	4	state	out	0	1	2	3	4
0	0	0	2	4	7	8	9	−5	1	2	3	14	12
1	−1	15	2	4	7	8	10	−6	1	2	4	5	1
2	−2	1	2	4	5	1	11	−6	1	2	4	7	8
3	−3	1	2	4	7	9	12	0	11	2	4	7	8
4	−3	1	2	4	6	12	13	0	11	2	4	7	9
5	−4	1	2	4	7	8	14	1	10	16	4	7	9
6	−4	1	2	4	7	9	15	−2	1	2	4	7	8
7	−4	1	2	3	13	12	16	−3	1	2	4	5	1
8	−5	1	2	4	7	8							

The automaton of Theorem C for $(1, 4)$ has 28 reachable states, and it minimises to the 17 states of Table 2.

Indexing conventions. Automaton sizes depend on the indexing. Throughout, "minimal" refers to the minimal msd-first DFAO that reads the base-τ expansion of $n \geq 0$, with leading zeros allowed, and outputs $d(n)$, where $d(0) = 0$. Fokkink and Joshi instead feed in $n - 1$, i.e. they describe $s(m) = d(m + 1)$, $m \geq 0$, up to an additive constant [6, Fig. 7]. An automaton for s is obtained from the one of Theorem C by letting S output $d(1)$ and letting each triple output its last coordinate. In that convention the minimal automata are smaller: 9 states for $(1, 4)$, and 6 states for $(1, 3)$, matching the 6-state automaton of [6, Fig. 7]. Table 3 lists both sizes.

Theorem 5.2. *Let $W(n)$ be the output of the DFAO in Table 2 on the base-5 expansion of n. Then $A_n = 11n + W(n)$ is the 4-rusty sequence. This statement is certified by Walnut, independently of Theorems A–C.*

Proof. In Walnut (v7.1) we define the predicates $\mathrm{ra}(n, s)$: $n > 0 \wedge s + 6 = 11n + \mathtt{RUS}[n]$; $\mathrm{rb2}(n, t)$: $\exists s\ \mathrm{ra}(n, s) \wedge (5t = s + 1 \vee 5t = s + 2)$; and $\mathrm{rb1}(n, u)$: $\exists s, t\ \mathrm{ra}(n, s) \wedge \mathrm{rb2}(n, t) \wedge u + 4t = s$. Walnut returns `TRUE` for each of the following: the initial values $\mathrm{rb1}(1, 1)$, $\mathrm{rb2}(1, 2)$ and $\mathrm{ra}(1, 9)$; totality of ra and rb2 for all $n > 0$; strict monotonicity of ra and the interleaving $B_1(n) < B_2(n) < B_1(n+1)$; disjointness of the range of ra from the ranges of rb1 and rb2; the cover statement that every $x > 0$ lies in one of the three ranges; the bounds $11n - 6 \leq A_n \leq 11n + 1$; and the statement that the first failure of $A_{5n+1} = 9 + 55n$ is at $n = 348$. Control queries, e.g. $\mathrm{ra}(1, 10)$, and a cover statement omitting ra, return `FALSE`. The relation $A_n = B_1(n) + 4B_2(n)$ holds by construction of rb1. The interleaving makes (B_m), defined by $B_{2n-1} = B_1(n)$ and $B_{2n} = B_2(n)$, strictly increasing. So (A, B) is a complementary pair of strictly increasing sequences satisfying (1), and by uniqueness [5, Lemma 3], [6, Lemma 2] it is the anti-recurrence pair of $(1, 4)$. A second, independently written query set gives the same verdicts (artifact files `walnut_ref14.*`). □

Walnut certificates of the same kind for $(1, 5)$, $(2, 7)$ and $(1, 1, 5)$ are described in Appendix B.

6. Related work and further directions

6.1. **Prior work.** We searched for any earlier proof of boundedness or automaticity for general linear forms in arXiv, zbMATH Open, the OEIS, Semantic Scholar, OpenAlex, DataCite, publisher pages and Google Scholar (October 10, 2026). Except where noted, we read each source listed here.

- *Kimberling–Moses* [12]: Theorems 2.1 and 3.2 determine growth rates for linear complementary equations; Section 6 introduces the anti-Fibonacci system and anti-m-nacci generalisations, with conjectured morphisms for the difference sequences; Example 12 conjectures bounded discrepancy for a related system. [13]: Theorem 1 gives existence and uniqueness for $a_n = f(b_{g(n)}, b_{h(n)})$, with explicit results for $a_n = b_{2n} + b_{4n}$; Section 6 studies $a_n = ub_n + vb_{n+k} + c$. Neither states a general boundedness or automaticity theorem for (1).
- *Bosma et al.* [3] and *Fokkink–Joshi* [5, 6]: see the introduction; [6, §6] states that the general case is open.
- *OEIS* [16, 17, 18, 19, 20]. The bounds in these entries are stated as conjectures, or have been proved for those specific forms only (by [3, 5]). The 3- and 4-rusty sequences are not in the OEIS.
- *Other complementary equations.* Kimberling [10, 11] treats families such as $b(n) = a(jn) \pm r$, $b(n) = a(jn) + kn$ and Wythoff-type equations. Shallit [21] proved complementarity statements for Tribonacci-based sequences in Walnut (see also [22]). Fraenkel [7], which [6] cites as an origin of complementary sequences, studies P-positions of generalised Wythoff games (we have seen only its opening section). Fraenkel and Peled [8] study pairs $a_n = \mathrm{mex}\{\dots\}$, $b_n = a_n + c_n$ and show (zbMATH summary; full text not seen) that membership is polynomial-time decidable when c_n is approximately linear. None of these treats (1).

No works citing [5, 6] were found, and searches of arXiv and zbMATH Open for "anti-recurrence", "anti-Fibonacci", "anti-k-nacci", "anti-bonacci" and "Clergyman" found nothing on anti-recurrences beyond the works cited here. We did not consult MathSciNet. To the best of our knowledge, boundedness and automaticity for general linear forms are new, as are the universal box, the two-term recursion and the explanation of the A_{1741} break.

6.2. **Background and comparison.** Counting arguments of the kind in Lemma 3.1(a) go back to Lambek and Moser [14]. Automatic sequences go back to Cobham [4]; see [1] for the theory. Since automatic and linear sequences are τ-regular and τ-regular sequences form a ring [2], Theorem C also shows that A_n itself is τ-regular.

Under A_1-boundedness, Fokkink and Joshi show that each interval $I_n = [\kappa(n-1)+1, \kappa n]$ contains exactly one A-term; the process is then a uniform substitution on residues, which they handle with Cobham's little theorem. Without A_1-boundedness this structure breaks down: the 0-transitions of the $(1,3)$ automaton [6, Fig. 7] all return to the initial state, which fails for $(1,4)$ [6, §5], and for $(1,4)$, d takes the value 1, so A_n can leave I_n. Our counting identity avoids intervals altogether: it reads $d(\tau q + r + 1)$ off the two values $d(q)$ and $d(q+1)$.

6.3. **Further directions.** Kimberling, partly with Moses, studied many systems of complementary equations [10, 11, 12, 13] and entered many sequences of this kind in the OEIS, often with empirical bounds stated as conjectures; A299409, A304499 and A304502 are examples [18, 19, 20]. Our results settle one class completely: for every linear anti-recurrence (1), $A_n - \kappa n$ is bounded and automatic, and Theorem C provides an explicit automaton from which further properties of a given form can be checked. Other classes remain. For example, for the equations $a_n = ub_n + vb_{n+k} + c$ with advanced subscripts, Kimberling and Moses report that tables and

experiments suggest that $a_n - (u+v+1)n$ is unbounded unless $u = v = 1$ [13, §6], so the linear-plus-automatic picture cannot be expected in general. Fokkink and Joshi ask, more broadly, which complementary sequences are sums of linear and automatic sequences [6, §6]. How far the counting method of Step 1 extends beyond (1) is open. Within the present class, Conjecture 3.5 and the observed lower bound $d \geq -K + 2$ (Remark 3.6(b)) remain open.

Data and code availability

All scripts, logs and Walnut files are in the reproducibility artifact accompanying this paper (with `README.md` and `SHA256SUMS`): Python generators and checkers, two separately written C generators with their checkers, the Walnut word automata `RUS`, `RFV`, `RTS`, `RQF` with command files and logs, and the scripts that build the automaton of Theorem C and compute exact ranges and automaton sizes. Walnut is free software [15]; we used version 7.1.

Use of AI tools

The proofs in this paper were produced with substantial AI assistance. AI agents working under the author's direction found and wrote the arguments of Sections 3–5, wrote all code (generators, checkers, automaton constructions and Walnut query files), ran the computations, carried out most of the literature search and drafted the text. The internal cross-checks and the separately written second implementation described in Appendix A were also carried out by AI agents; they are not peer review. The general proofs of Sections 3 and 4 are not machine-checked; the Walnut certificates of Theorem 5.2 and Appendix B are: they were produced by Walnut itself and can be re-run from the artifact. R. Fokkink read an earlier draft and found the argument convincing; this was not a formal refereeing. The author initiated and guided this experiment but has not independently verified the mathematics; readers should treat the general proofs as unrefereed until checked by experts.

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Appendix A. Computational verification

None of the following is needed for the proofs; it guards against slips in the hand arguments. Every check generates (A, B) by brute force from the definition (mex plus linear form), independently of the formulas being tested, on a single CPU core under a 500 MB memory cap. No check reported a failure.

(1) *Large-scale check for* $(1, 4)$. For 10^7 terms, $d \in [-6, 1]$ and (6) holds for every $n \leq 10^7$. The 17-state DFAO agrees with $d(n)$ for every $n \leq 2 \cdot 10^6$.

(2) *Step-by-step recheck on* 50 *forms* (20 not A_1-bounded), $6 \cdot 10^4$ terms each: $B_m = m + c(m)$ and identity (3); conditions (i)–(iv) for the explicit box and the pointwise bounds $\Phi(r, U) \leq d(n) \leq \Phi(r, L)$; the window recursion of Appendix C.

(3) *An independently written second implementation (*585 *forms).* This and item 5 use a separately written C generator; they are not peer review. Forms: all $k = 2$ with $a_j \leq 12$; all $k = 3$ with $a_j \leq 6$; 25 adversarial forms; 200 random forms with $k \in \{3, 4, 5\}$ and $a_j \leq 12$ (178 forms in total are not A_1-bounded). Up to 10^5 terms per form, $2 \cdot 10^6$ for 8 forms. Checked: the identity, the box, the window recursion, and determinism of the window map.

(4) *Checks of the present formulation.* (a) 433 forms (all $k = 2$ with $a_j \le 12$, all $k = 3$ with $a_j \le 5$, 150 random forms with $k \in \{3, 4, 5\}$ and $a_j \le 12$, and the extremal forms $(1, \dots, 1)$, $(1, \dots, 1, 20)$, $(20, 1, \dots, 1)$), 6000 terms each: Lemma 3.4, $U \le \tau - k$, and Theorem B for all $q \ge 0$. (b) 17 forms, $2 \cdot 10^4$ terms: the explicit constants of Proposition 3.3 and Appendix C, coordinate by coordinate. (c) 12 named forms: the automaton of Theorem C reproduces $d(0), \dots, d(2 \cdot 10^4)$ exactly; after minimisation it has 17, 19, 37 and 24 states for $(1, 4)$, $(1, 5)$, $(2, 7)$ and $(1, 1, 5)$, matching the Walnut-certified automata of Appendix B. (d) The 10 named and 736 further forms of Section 3.3: exact ranges from the reachable states of the automaton of Theorem C, narrowest boxes, the excess of (L^*, U^*), $V(1) \in [-K, \tau - k]^3$, and minimal sizes in both conventions.

(5) *A second pass of the separately written implementation (*2270 *forms*, $2 \le k \le 8$, entries up to 200, τ up to 1600): Lemma 3.4, Theorem B for all $q \ge 0$, the maps H_c of (5), and that the reachable triples of Theorem C lie in $[-K, \tau - k]^3$. Also $3 \cdot 10^6$ terms for 7 large forms; for $(1, 4)$, $A_{1741} = 19148$ and $A_{5q+2} = 19 + 55q$ for all $5q + 2 \le 2 \cdot 10^6$; and an exact rational recheck of the constants of Proposition 3.3 and Appendix C on 1420 forms.

Appendix B. Further Walnut certificates

For the forms in Table 3, a DFAO was built from the window recursion and minimised, its outputs were stored as a Walnut word automaton, and the analogue of the query set in Theorem 5.2 was run: initial block, totality, monotonicity of A, interleaving of the B's, disjointness of A from the B's, cover of $\mathbb{N}$, and the linear relation, plus control queries. All verdicts were as expected. The B's are pairwise disjoint because of the interleaving query $B_1(n) < \cdots < B_k(n) < B_1(n+1)$; for $(1, 1, 5)$ the separately written implementation also ran the three pairwise disjointness queries for B_1, B_2, B_3 directly, with its own predicate definitions, and all return `TRUE` (artifact files `walnut_rqf_disjB.*`). For $(1, 1, 5)$ the direct query for the linear relation exceeded a 400 MB heap; it was replaced by an equivalent reformulation that first defines the sum $B_1 + B_2 + 5B_3$ as a predicate, and both resulting queries return `TRUE`. For $(1, 5)$, $(2, 7)$ and $(1, 1, 5)$ the relation is checked by an explicit Walnut query, not just by construction. None of the forms is A_1-bounded. The certificates are not needed given Theorem C; they are an independent check of it.

Table 3. Walnut-certified forms (Walnut v7.1). "States" is the size of the minimal msd-first DFAO in our convention (input $n \ge 0$, output $d(n)$, $d(0) = 0$); "FJ" is the size in the convention of [6] (input $n - 1$). The range of d is exact.

a	τ	κ	states	FJ	range of d	Walnut automaton
$(1, 4)$	5	11	17	9	$[-6, 1]$	`RUS`
$(1, 5)$	6	13	19	11	$[-7, 1]$	`RFV`
$(2, 7)$	9	19	37	20	$[-12, 3]$	`RTS`
$(1, 1, 5)$	7	22	24	9	$[-9, -1]$	`RQF`

Appendix C. A version of Step 2 with an arbitrary box

Let $[L, U]$ be any box satisfying (i)–(iv). Let $T = \max(1, \lceil U/K \rceil)$, and let T' be the least integer with $KT' > K - 1 - L$. By (3), for $q \ge T$,

$$d(\tau q + r + 1) = A_1 - K - 1 - r - \tau T + \sum_j a_j \sum_{t=-T+1}^{T'-1} [\, d(q + t) \le u_j - Kt \,].$$

Put $o = -2T$, $W = 2T + 2T' + 1$, $P_0 = 2T + 1$, and $V(p) = (d(p+o), \ldots, d(p+o+W-1))$. Then $o \le 0 \le o + W - 1$, and the containment conditions

$$\lfloor (o-1)/\tau \rfloor - T + 1 \ge o, \qquad \lfloor (\tau + o + W - 3)/\tau \rfloor + T' - 1 \le o + W - 1$$

hold for all $\tau \ge 2$. Hence $V(\tau p + c) = H_c(V(p))$ for $p \ge P_0$, with maps H_c on $[L, U]^W$ (the *window recursion*). Iterating, with $b = \sum_{i<e} c_i \tau^i$ (c_0 the least significant digit),

$$V(\tau^e p + b) = H_{c_0} \circ H_{c_1} \circ \cdots \circ H_{c_{e-1}}(V(p)) \qquad (p \ge P_0).$$

So each kernel sequence $p \mapsto d(\tau^e p + b)$ is determined by finitely many data: the map $\pi \circ H_{c_0} \circ \cdots \circ H_{c_{e-1}}$, where π is the coordinate at offset 0, and its first P_0 values. The τ-kernel is therefore finite, and the kernel characterisation [1, Ch. 6] applies. With the universal box, $T = 1$ and $T' = 2$.

References

[1] J.-P. Allouche and J. Shallit, *Automatic Sequences: Theory, Applications, Generalizations*, Cambridge University Press, 2003. doi:10.1017/CBO9780511546563.

[2] J.-P. Allouche and J. Shallit, The ring of k-regular sequences, *Theoret. Comput. Sci.* **98** (1992), 163–197. doi:10.1016/0304-3975(92)90001-V.

[3] W. Bosma, R. Bruin, R. Fokkink, J. Grube, A. Reuijl, and T. Tromp, Using Walnut to solve problems from the OEIS, arXiv:2503.04122 [math.NT], 2025. Published as J. Integer Seq. **28** (2025), no. 3, Article 25.3.8.

[4] A. Cobham, Uniform tag sequences, *Math. Systems Theory* **6** (1972), 164–192. doi:10.1007/BF01706087.

[5] R. Fokkink and G. Joshi, Anti-recurrence sequences, arXiv:2506.13337v1 [math.NT], 2025.

[6] R. Fokkink and G. Joshi, Anti-recurrence sequences, *Integers* **26** (2026), #A24. doi:10.5281/zenodo.18305140.

[7] A. S. Fraenkel, How to beat your Wythoff games' opponent on three fronts, *Amer. Math. Monthly* **89** (1982), no. 6, 353–361. doi:10.1080/00029890.1982.11995454.

[8] A. S. Fraenkel and U. Peled, Harnessing the unwieldy MEX function, in: R. J. Nowakowski (ed.), *Games of No Chance 4*, Cambridge University Press, 2015, pp. 77–94. doi:10.1017/9780511820809.008.

[9] D. R. Hofstadter, Anti-Fibonacci numbers, unpublished note, October 2014. Available at `https://oeis.org/A075326/a075326_1.pdf`.

[10] C. Kimberling, Complementary equations, *J. Integer Seq.* **10** (2007), Article 07.1.4.

[11] C. Kimberling, Complementary equations and Wythoff sequences, *J. Integer Seq.* **11** (2008), Article 08.3.3.

[12] C. Kimberling and P. J. C. Moses, Linear complementary equations and systems, *Fibonacci Quart.* **57** (2019), no. 5, 96–110.

[13] C. Kimberling and P. J. C. Moses, Complementary equations with advanced subscripts, *J. Integer Seq.* **24** (2021), Article 21.3.3.

[14] J. Lambek and L. Moser, Inverse and complementary sequences of natural numbers, *Amer. Math. Monthly* **61** (1954), no. 7, 454–458. doi:10.1080/00029890.1954.11988496.

[15] H. Mousavi, Automatic theorem proving in Walnut, arXiv:1603.06017, 2016.

[16] OEIS Foundation Inc., The On-Line Encyclopedia of Integer Sequences, Sequence A075326 (Anti-Fibonacci numbers), `https://oeis.org/A075326`; see also A249031, A249032.

[17] OEIS, Sequence A265389, `https://oeis.org/A265389`.

[18] OEIS, Sequence A299409, `https://oeis.org/A299409`.

[19] OEIS, Sequence A304499, `https://oeis.org/A304499`.

[20] OEIS, Sequence A304502, `https://oeis.org/A304502`.

[21] J. Shallit, Some Tribonacci conjectures, *Fibonacci Quart.* **61** (2023), no. 3, 214–221. doi:10.1080/00150517.2023.12427399.

[22] J. Shallit, *The Logical Approach to Automatic Sequences: Exploring Combinatorics on Words with Walnut*, Cambridge University Press, 2022. doi:10.1017/9781108775267.

[23] T. Zaslavsky, Anti-Fibonacci numbers: a formula, unpublished note, September 26, 2016. Available at `https://oeis.org/A075326/a075326_2.pdf`.